

Reactive H₂S Stripping Column Simulation Software with CO₂ Hydration Kinetics

Motivation & Significance

- Turns waste Na₂SO₄ into Na₂CO₃
- CO₂ hydration is the rate-limiting step
- Model includes real kinetics

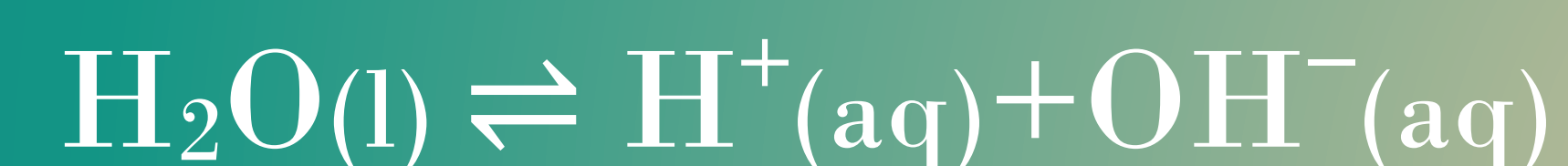
The Chemistry

CO₂ Hydration:



This is the rate limiting step in the reactive stripping process with $k_f = 0.03 \text{ s}^{-1}$

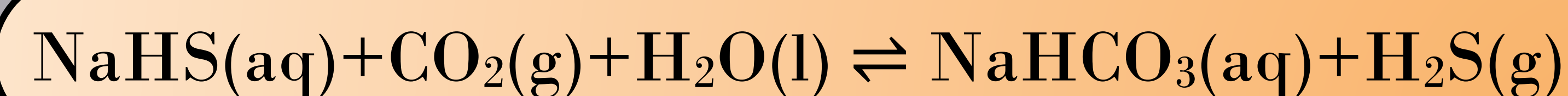
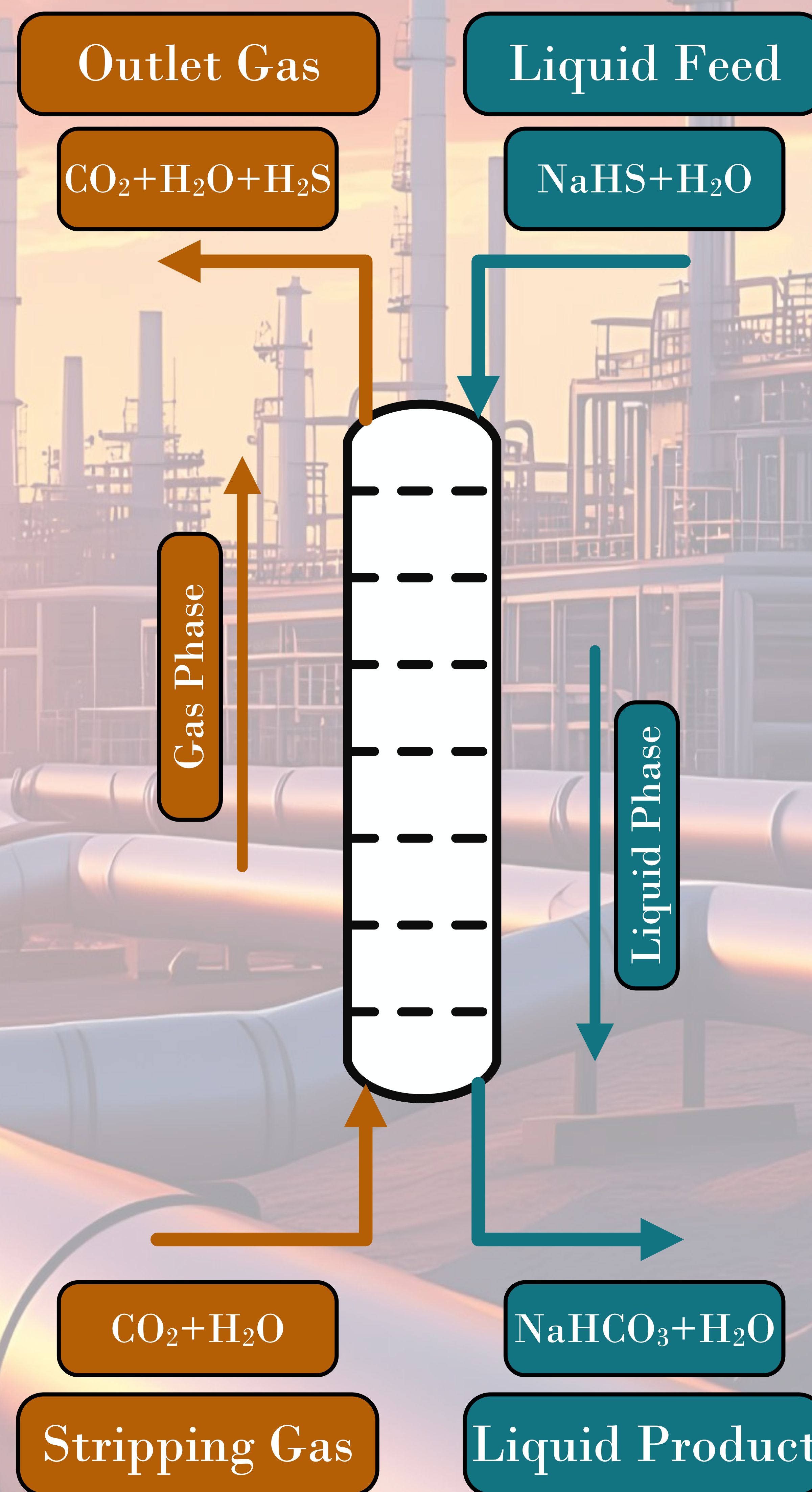
Important Equilibria:



Materials & Methodology

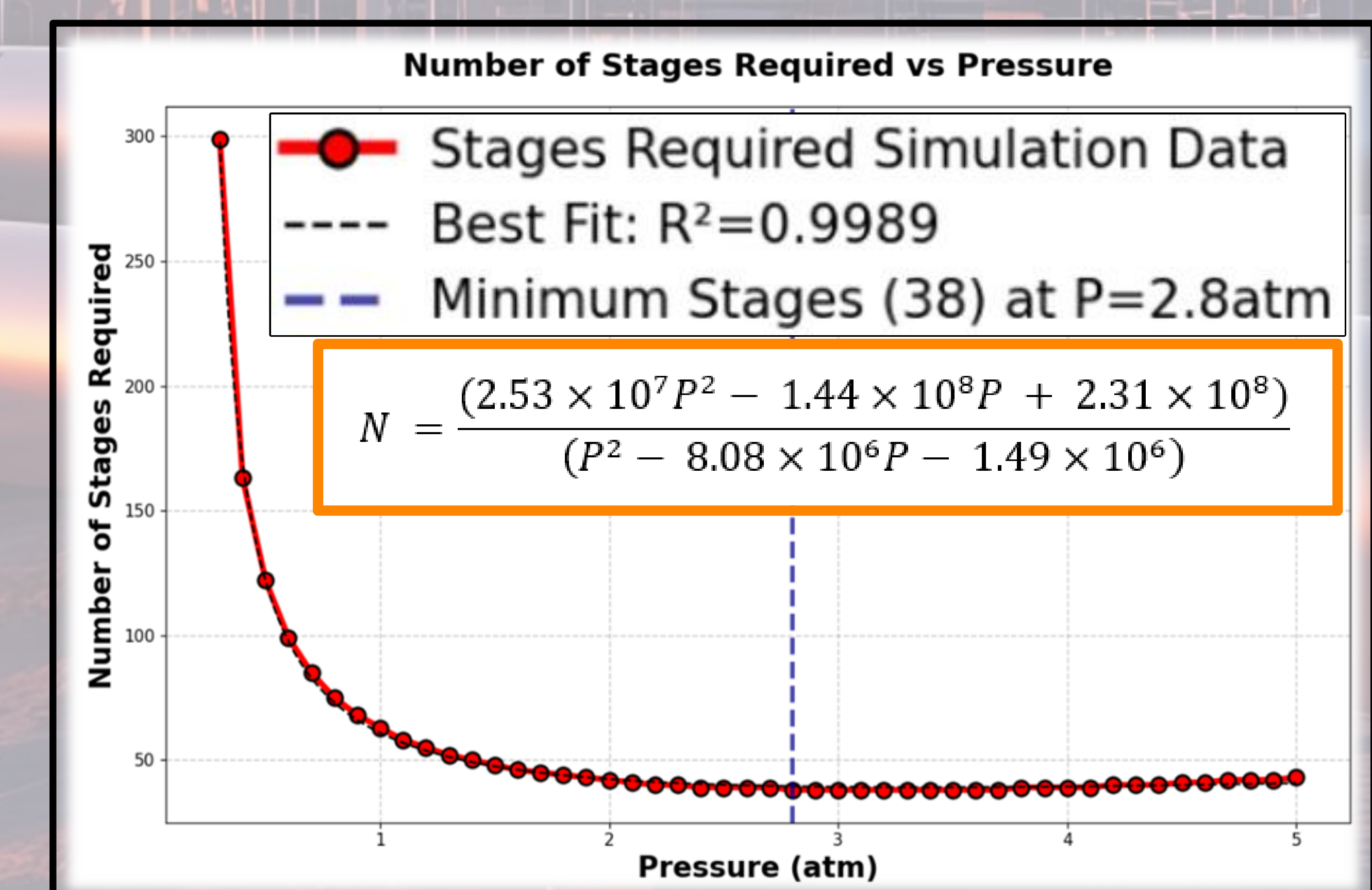
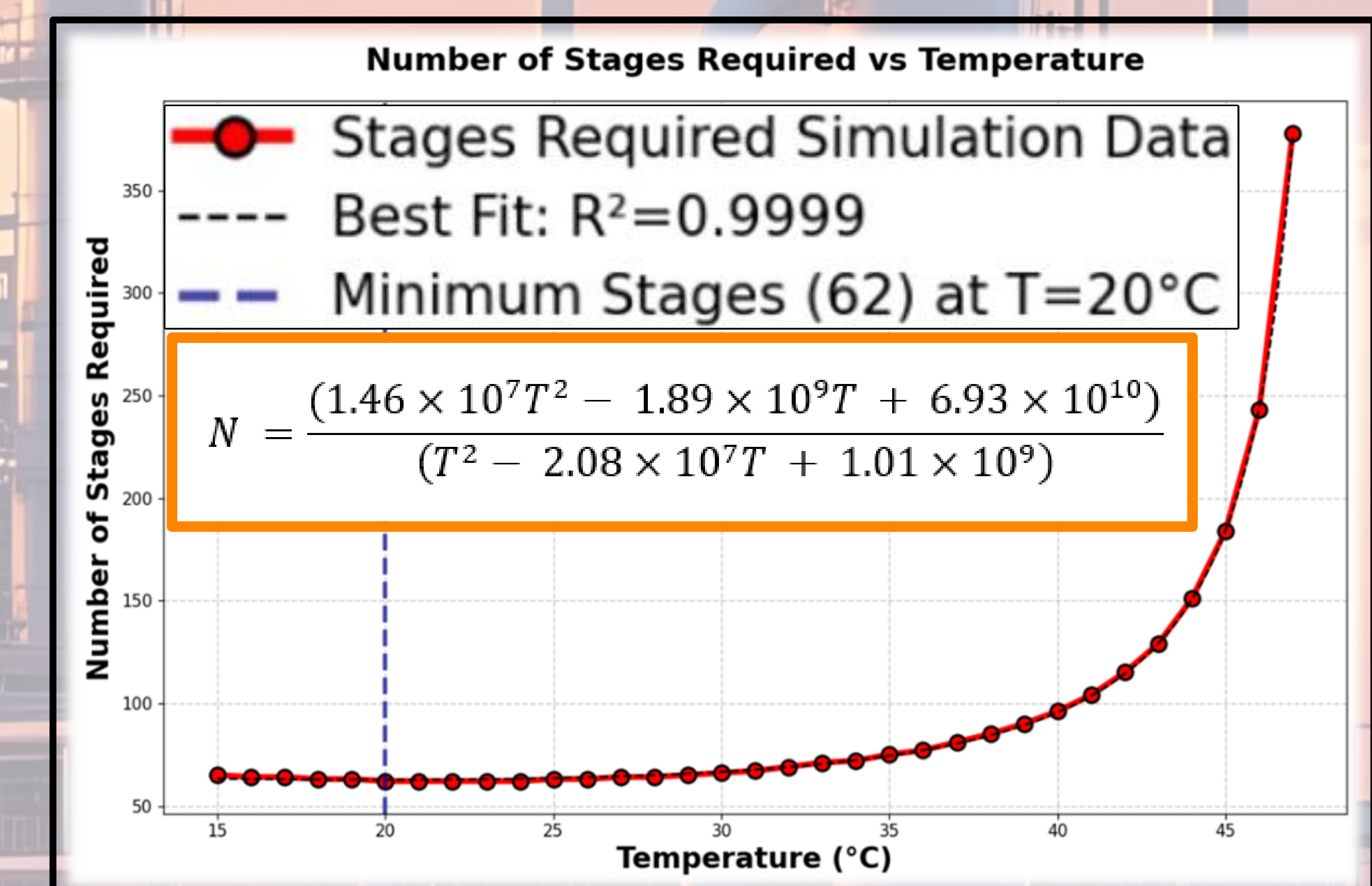
- Stage-wise Python model
- Includes kinetics and equilibria
- Solved with fsolve
- Modular and automated software
- Runs sensitivity analyses
- Software generated visuals

The Model:



Results & Analyses

- Kinetics limit high H₂S recovery
- Best at 20 °C, 2.8 atm
- Equilibrium models overpredict



Impact & Conclusions

- Improves design accuracy
- Enables waste valorization
- Guides industrial use

Poster by:
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Online link to my article:



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